The calibration epoch as the unit of evidence: preregistered parity-leakage and layout-transfer experiments on IQM Garnet

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We report five preregistered experiments on the IQM Garnet 20-qubit square-lattice processor, each with its decision rule committed before any hardware data existed and each reported regardless of outcome. First, the even-over-odd parity-sector leakage asymmetry previously measured on IBM Heron r2 replicates on Garnet at preregistered power (relative asymmetry +0.068, one-sided $p = 1.1 \times 10^{-5}$). Because the noiseless parity leakage of these number-conserving XY-Trotter circuits is exactly zero, this is a statement about device noise. Second, the asymmetry is not stable: over ten execution windows it varies beyond shot noise at fixed depth (Cochran $Q = 19.01$ on 9 degrees of freedom, $p = 0.025$), more strongly when windows are pooled by provider calibration set ($p = 0.0083$), and an isolated sign reversal at depth 10 does not replicate. Third, a confirmatory campaign that takes the calibration epoch as its independent unit — twelve epochs, 688,128 shots — finds the asymmetry larger at depth 8 than at depth 12 (mean contrast +0.0173, one-sided $p = 0.026$); the lower bound is +0.0030 and three of twelve epochs have the opposite sign. Fourth, a calibration-aware layout optimiser developed on heavy-hex devices fails its frozen transfer rule: it is better than automatic placement at chain lengths 8 and 16 and worse at 12, and a powered follow-up confirms the reversal. All preregistrations, calibration snapshots, job identifiers, raw counts and analysis scripts are public.

I. MOTIVATION AND PRIOR STATE

Two threads of our published programme meet on a square lattice.

First, the **layout thread**: our fidelity-aware placement stack — a community-detection analysis pass, a Kuramoto-XY-aware discrete layout cost model, and a multi-restart discrete optimiser — was built and benchmarked entirely on IBM heavy-hex devices. Whether calibration-aware placement gains survive a change of lattice topology is an open, publishable question in both directions, and square-lattice Garnet is the natural second data point.

Second, the **parity thread**: on IBM Heron r2 (`ibm_fez`, `ibm_kingston`) we measured a robust even-over-odd parity-sector leakage asymmetry in number-conserving Kuramoto-XY [1, 2] Trotter circuits (+10.8 % ± 1.1 pooled, Fisher $p \ll 10^{-16}$) [3]. An exact statevector baseline subsequently proved the noiseless parity leakage of these circuits is exactly zero at every depth (excitation-number conservation), reframing the result: the asymmetry is a property of device noise, not of coherent dynamics, and the live question became whether that noise structure is backend-universal [4]. A 2026-05-13 Garnet pilot (six layouts, 12,288 shots) was too weak to answer: combined p between 0.05 and 0.57, sign agreement 1/3 to 3/3 across layouts. Only layout [2, 7, 12, 13] agreed with IBM in sign at all three depths — which is why the present confirmatory block pins that layout, discloses the selection, and treats the May data as design input, never as evidence.

II. METHODS COMMON TO BOTH EXPERIMENTS

Hardware access. All quantum-hardware results in this work were obtained on IQM Garnet through **IQM Resonance**; we thank IQM for extended Resonance access. The two core experiments ran on 20 qubits of the square-lattice device on 2026-07-21 (calibration set `246a4930-54e3-4cd9-a2d1-fcc0919675f5`); explicitly identified follow-ups used fresh windows and their own recorded calibration sets. The window-variability campaign (Sec. VI) ran in ten windows between 2026-07-22 and 2026-09-04, and the depth-ordering confirmation (Sec. VII) in twelve calibration epochs between 2026-09-23 and 2026-10-06, each under its own recorded calibration set.

Device and access. IQM Garnet (20 qubits, square lattice) [5] via IQM Resonance, 2026-07-21, single calibration window. Live calibration was snapshotted through the published quality-metrics endpoint before circuit freezing (per-edge CZ gate fidelity; per-qubit readout error), committed as a JSON artefact, and used both to drive the optimiser arm of Experiment 1 and to verify layout availability for Experiment 2.

Preregistration. Both campaign documents were committed before execution (`docs/campaigns/iqm_layout_transfer_square_lattice_prereg_2026-07-21.md`, `docs/campaigns/iqm_dla_backend_sensitivity_powered_prereg_2026-07-21.md`), freezing hypotheses,

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endpoints, shot budgets, stop rules, and decision rules. One pre-submission amendment was made to Experiment 1 (Sec. III A) before any hardware data existed; the original text and the evidence that forced the amendment are retained in the document.

Readiness gates. Neither experiment was submitted before passing: unit tests to the repository standard (100 % line and branch coverage on the new harness modules), a complete `IQMFakeGarnet` dry run of every circuit from committed code, committed exact classical baselines, and — for Experiment 1 — a transpiled two-qubit depth-parity gate across arms; for Experiment 2 — a transpiled depth envelope (May reference depth 159 + 25 %). A zero-spend server-side `garnet:mock` submission validated the full submit–retrieve loop before each first real block.

Spend discipline. Blocks were submitted incrementally, with dashboard credit checks between blocks (the first block of each experiment carried a preregistered abort threshold). The entire dual campaign consumed roughly 13 of 30 monthly credits.

III. EXPERIMENT 1 — LAYOUT-OPTIMISER TRANSFER TO THE SQUARE LATTICE

A. Design

Identical logical circuits per size: two-edge-colour width-2 XY-Trotter chains (5 Trotter steps = 20 two-qubit layers, constant in n), initial state at quarter filling (`1000` repeated), which fixes the conserved mean-Z order-parameter proxy at exactly $R = 0.5$ for every size and depth. The endpoint, $\mathrm{err}(\mathrm{arm}, n) = |R_{\mathrm{hw}} - 0.5|$, uses the absolute mean Z-magnetisation proxy — a sum of single-qubit marginals, so the two readout-calibration circuits per size (all-zeros, all-ones over the union of measured qubits) support an exact tensored per-qubit readout correction for this observable; no full-matrix mitigation is claimed.

Three arms per size, identical circuits, different placements:

1. **optimised** — highest-fidelity connected chain region from the live calibration snapshot, order polished by the discrete Kuramoto layout optimiser under a deterministic, transpiler-free chain depth model;
2. **default** — Qiskit [6] automatic placement (optimisation level 1, frozen transpiler seed);
3. **naive** — calibration-blind topological baseline (Amendment 1): the lexicographically smallest connected chain on the coupling graph.

Amendment 1 (pre-submission). The originally preregistered naive arm — consecutive physical indices (0, . . . , n−1) — is not a connected path on Garnet; the fake-backend readiness run showed routing SWAPs inflating its two-qubit depth to 75/110/83 versus 40 for the other arms, violating the frozen depth-parity validity gate at every size. The arm was redefined (calibration-blind, topology-only) before any hardware data existed.

n	optimised	default	naive	optimised wins?
8	**0.1475**	0.2533	0.2971	yes
12	0.2088	**0.1846**	0.4138	**no**
16	**0.2924**	0.3144	0.3378	yes

TABLE I. Experiment 1: readout-corrected order-parameter error against the exact reference $R = 0.5$ for the three placements. Bold marks the smaller error of the optimised and default arms.

After amendment, all three arms transpile to identical two-qubit depth (40) at every size, so fidelity differences are attributable to *which* qubits and edges are used, not to circuit volume.

Matrix: 3 sizes × 3 arms × 2,048 shots + 3 × 2 readout states × 1,024 shots = 15 circuits, 24,576 + 6,144 shots. On the live calibration the three arms separated onto genuinely different regions at $n = 8$: optimised (16, 11, 10, 15, 14, 13, 17, 18), default (1, 0, 3, 2, 7, 12, 13, 17), naive (0, 1, 4, 3, 2, 7, 8, 9).

Frozen primary rule. err(optimised) < err(default) at **all three** sizes AND the bootstrap 90 % CI of the pooled error difference (10,000 shot-level multinomial resamples of every circuit including the readout calibrations, seed 20260721) excludes zero. Secondary: err(default) versus err(naive); two-qubit depth parity within 10 % as a validity gate.

B. Results

Table I lists the readout-corrected order-parameter error versus the exact reference $R = 0.5$ (jobs `019f8639-96be...`, `019f8639-980e...`, `019f8642-e162...`, `019f8642-e326...`, `019f8643-650c...`, `019f8643-6712...`).

Pooled err(default) − err(optimised) = **+0.0345**, bootstrap CI90 [+0.0260, +0.0431]. Pooled err(naive) − err(default) = **+0.0989**, CI90 [+0.0898, +0.1079]. Depth parity held exactly (40/40/40 at every size). Figure 1 shows both runs.

C. Interpretation (bounded, as preregistered)

The frozen primary rule fails. The all-sizes condition breaks at $n = 12$, so no transfer claim is made — even though the pooled interval excludes zero. We report exactly the preregistered bounded conclusion:

- calibration-aware placement carries a real pooled advantage on this device and calibration window, but the advantage is **not uniform across sizes**; the automatic transpiler matched or beat it at one of three sizes;
- **placement itself matters strongly** on the square lattice: the calibration-blind topological baseline is worse than automatic placement at every size, by three times the optimiser's pooled margin.

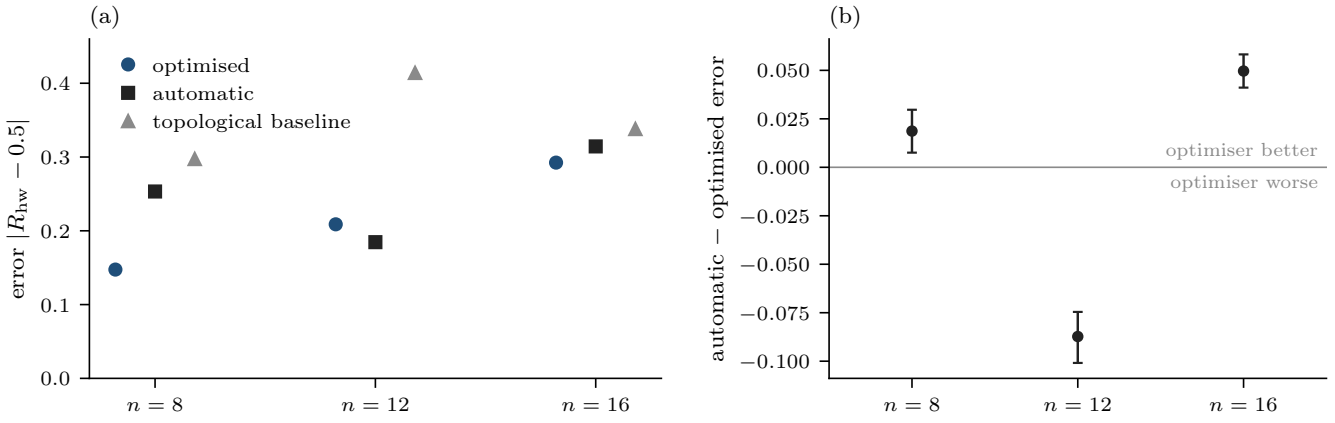


FIG. 1. Layout transfer on Garnet. (a) Readout-corrected order-parameter error of the three placements in the first experiment (2,048 shots per point). (b) Powered follow-up: corrected error of the automatic placement minus that of the optimised placement, with bootstrap 95 % intervals (8,192 shots per arm). The optimised placement is better at $n = 8$ and $n = 16$ and worse at $n = 12$ in both runs.

A transfer claim therefore required a fresh design powered for per-size conclusions and uniformity. That preregistered follow-up has now executed.

D. Powered per-size layout-transfer follow-up

The per-size layout-transfer study was frozen on 2026-07-22 before its hardware data existed (`docs/campaigns/iqm_layout_transfer_per_size_prereg_2026-07-22.md`). It repeated every arm four times at each size (8,192 main shots per arm/size), added a fresh all-zero/all-one readout pair per size, and submitted the whole 42-circuit, 79,872-shot matrix in one pass on 2026-07-26 under calibration set `c2097be4-1e23-49bc-adaa-8e8c01df6223`. Provider transpilation preserved an identical two-qubit depth of 40 for every main circuit.

The frozen corrected default − optimised endpoint was significant at all three sizes after Holm correction [7], but not in one direction (Table II).

The powered design therefore confirms, rather than resolves away, the $n = 12$ reversal. Cochran's heterogeneity statistic [8] rejects a common effect across sizes ($Q = 292.43$ on 2 df, $p = 3.16 \times 10^{-64}$). Raw-count sensitivity preserves all three directions. The pooled difference is slightly negative (-0.00632, CI90 [-0.01187, -0.00099]) because the $n = 12$ disadvantage dominates; it must not be used to erase the per-size result. This is a size-dependent layout effect in one Garnet calibration window, not a quantum-advantage, backend-universal, or calibration-aware-superiority claim.

IV. EXPERIMENT 2 — PARITY-SECTOR NOISE ASYMMETRY AT PREREGISTERED POWER

A. Design

Committed May-campaign builders, bit-identical circuits: $n = 4$ XY-Trotter chains, states `0011` (even sector) and `0001` (odd sector), depths 4, 6, 10, four repetitions per state/depth at 1,024 shots (calibration-drift resolution), plus one four-state readout block (`0011`, `0001`, `0000`, `1111` at 2,048 shots) — 28 circuits, 32,768 shots, all on the pinned layout [2, 7, 12, 13] (fallback [9, 4, 3, 8] was not needed; all three layout edges were calibrated, CZ fidelities 0.982–0.990). Transpiled depths at submit time: 69 / 99 / 159 for depths 4 / 6 / 10 — the deepest exactly reproducing the May envelope reference of 159, within the frozen +25 % bound of 198.

Frozen primary. Pooled across depths and repetitions (12,288 shots per arm), one-sided two-proportion z-test of $\mathrm{leak}_{\mathrm{even}} > \mathrm{leak}_{\mathrm{odd}}$ at $\alpha = 0.05$; committed power analysis: 90 % power at +0.044 relative asymmetry (the weakest IBM per-depth reference). Secondary: per-depth Wilson intervals and one-sided tests, per-repetition drift table, sign agreement with the IBM direction. Claim boundary: with noiseless leakage exactly zero, every statement is about device noise; no coherent-dynamics claim is available.

B. Results

Jobs `019f866c-9c99...`, `019f866c-9f20...`, `019f8677-33b7...`, `019f8677-7a0b...`, `019f8677-ca9a...`

Primary: H0 rejected. Pooled $\mathrm{leak}_{\mathrm{even}} = 0.4194 > \mathrm{leak}_{\mathrm{odd}} = 0.3928$; relative asymmetry $+\mathbf{0.0677}$; $z = 4.247$; one-sided $p = 1.1 \times 10^{-5}$. Garnet shows a positive even-over-odd parity-sector leakage asymmetry in the

n	default − optimised	bootstrap CI95	Holm p	direction
8	+0.01868	[+0.00755, +0.02968]	0.001200	optimiser advantage
12	**−0.08728**	[−0.10090, −0.07459]	0.000600	**optimiser disadvantage**
16	+0.04964	[+0.04109, +0.05823]	0.000600	optimiser advantage

TABLE II. Powered per-size follow-up: corrected default-minus-optimised error with bootstrap 95 % intervals and Holm-adjusted p values.

IBM direction at preregistered power.

Secondary depth profile. Disclosed, not preregistered as an endpoint (Table III).

C. Interpretation

The pooled IBM-direction asymmetry replicating on a second superconducting architecture — square-lattice Garnet after heavy-hex Heron r2 — is evidence against the asymmetry being an artefact of one device family. To be precise about what the two devices share and what they do not: **shared** is the pooled even-over-odd sign of the parity-sector leakage asymmetry at statistical power; **divergent** is the depth dependence (flat on IBM, decaying and sign-reversing on Garnet), which may well reflect a different underlying mechanism. "Replication" here therefore binds only to the pooled directional effect, not to a claim that the two devices exhibit the same noise structure.

Put side by side (Fig. 2), the two devices differ first in scale. On `ibm_kingston` (the Phase 2 reference: 30 repetitions per sector and depth) the parity leakage of this circuit family rises from 0.08 at depth 2 to 0.28–0.29 at depth 20, and the even-minus-odd difference stays between +0.005 and +0.010 from depth 4 to depth 20. On Garnet the leakage is already 0.31–0.38 at depth 4 and reaches 0.48–0.50, the value of a fully mixed parity, at depth 12; the difference is +0.071 at depth 4 in the July window and shrinks as the leakage saturates. Garnet thus reaches within twelve Trotter steps a regime that Heron r2 has not reached at twenty. The devices share the sign of the asymmetry and differ in its size and in its dependence on depth. The comparison is descriptive: different days, different numbers of repetitions, and not a preregistered endpoint.

The initial secondary profile was qualitatively different from IBM's: IBM's reference asymmetry was approximately flat in depth ($\approx +0.044$ to $+0.087$), while the first Garnet window fell from +0.230 at depth 4 to −0.049 at depth 10. That isolated negative point initially motivated a zero-crossing hypothesis, but it was not a preregistered endpoint. The preregistered follow-ups in Sec. V do not support that hypothesis: the depth-8 versus depth-12 ordering is a bounded null ($p = 0.064$), a powered fresh-window depth-10 retest is positive, and the two depth-10 windows differ significantly ($p \approx 0.006$). The surviving bounded interpretation is therefore window-to-window variability of device-noise asymmetry, not a depth-localised zero crossing. No zero-crossing or equilibration-mechanism claim is made.

V. PREREGISTERED DEPTH-PROFILE FOLLOW-UP

A depth-profile campaign (depths 8 and 12, same layout, states, builders, and shot discipline; 20 circuits, 24,576 shots) was preregistered separately (`docs/campaigns/iqm_dla_depth_profile_prereg_2026-07-22.md`) — frozen before any of its data existed — and executed on 2026-07-22 under the same calibration set as Experiment 2 (jobs `019f8772-6338...`, `019f8772-665d...`, `019f87eb-30b1...`, `019f87eb-f2f8...`, `019f87ec-81b0...`).

Its frozen primary (decay ordering) fails to reject the null: $\Delta_8 = +0.0369$ exceeds $\Delta_{12} = +0.0132$ in the hypothesised direction, but the one-sided test gives $z = 1.52$, $p = 0.064 > 0.05$, so the decay ordering is not established at the committed power and we report the bounded null with the same prominence as any positive result.

Table IV gives the secondary, descriptive structure across the joint profile (depths 4/6/10 from Experiment 2's execution window, 8/12 from this one — a cross-window comparison frozen into the preregistration, mitigated by the shared calibration set).

Two hypothesis-generating observations arose: the mean total leakage rises monotonically toward 0.5, consistent with sector equilibration compressing the asymmetry at depth; and the sign pattern (+, +, +, −, +) places the depth-10 negative between positive neighbours, suggesting a window fluctuation rather than a systematic zero crossing.

A third preregistered campaign tested the second reading directly (`docs/campaigns/iqm_dla_d10_retest_prereg_2026-07-22.md`, frozen before its data; 8,192 shots per arm — 90 % power at the observed −0.0237 effect, which the depth-profile arms could not resolve; jobs `019f8809-42d7...`, `019f8809-448d...`). **The depth-10 negative sign does not replicate:** in the fresh window the depth-10 asymmetry is positive ($\text{leak}_{\text{even}}$ 0.4611 > leak_{odd} 0.4478; +0.030 relative), the one-sided test for the negative sign gives $p = 0.956$, and the preregistered cross-window secondary finds the two windows' depth-10 asymmetries genuinely differ ($z = 2.74$, two-sided $p \approx 0.006$). Per the frozen wording this strengthens — without proving — the window-fluctuation reading; with the fresh-window point positive, every depth in the joint profile carries a positive sign, and the surviving open

depth	$\text{leak}_{\text{even}}$	leak_{odd}	relative asymmetry	one-sided p	sign = IBM?
4	0.3811	0.3098	**+0.230**	5.8×10^{-12}	yes
6	0.4197	0.3875	**+0.083**	1.5×10^{-3}	yes
10	0.4575	0.4812	**−0.049**	0.98	**no**

TABLE III. Experiment 2: secondary depth profile of the parity-sector leakage. Disclosed, not a preregistered endpoint.

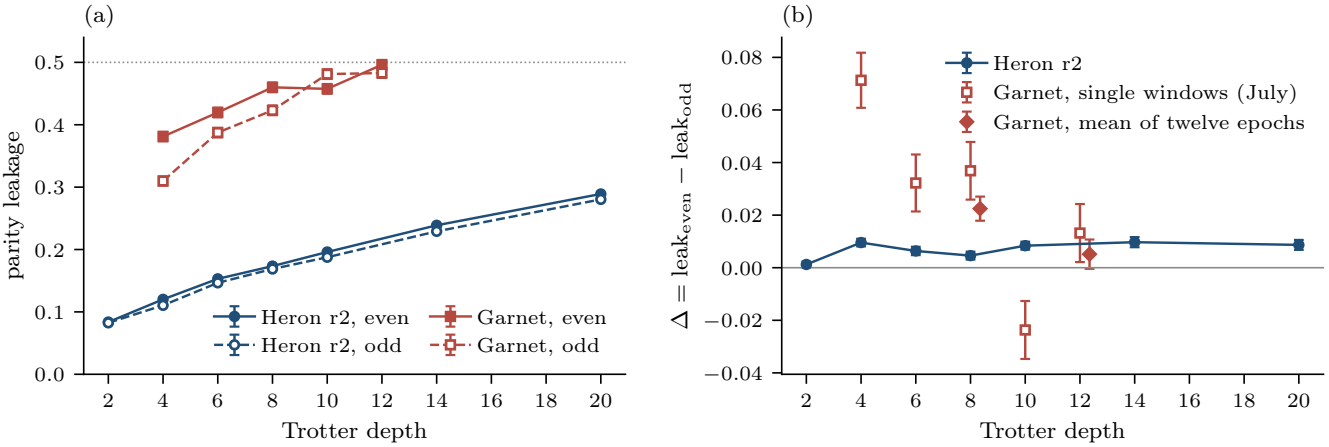


FIG. 2. Parity-sector leakage of the four-qubit XY-Trotter circuits on IBM Heron r2 (`ibm_kingston`, Phase 2 reference) and on IQM Garnet. (a) Leakage of the even and odd sectors; the dotted line is the fully mixed value. (b) Even-minus-odd difference. Open squares: single Garnet windows of July 2026 (4,096 shots per arm, binomial standard errors). Diamonds: random-effects means over the twelve calibration epochs of Sec. VII, offset horizontally for legibility. Heron r2 error bars are standard errors over 30 repetitions. Descriptive comparison, not a preregistered endpoint.

depth	relative asymmetry	mean total leakage	sign
4	+0.230	0.345	+
6	+0.083	0.404	+
8	+0.087	0.442	+
10	−0.049	0.469	−
12	+0.027	0.489	+

TABLE IV. Joint depth profile: depths 4, 6 and 10 from the execution window of Experiment 2, depths 8 and 12 from the depth-profile follow-up under the same calibration set.

question is the size of the window-to-window variability of the asymmetry at fixed depth, not a zero crossing.

VI. WINDOW-TO-WINDOW VARIABILITY OF THE ASYMMETRY

Sec. V left one question: how large is the window-to-window variability of the parity-sector asymmetry at a fixed depth. A fourth campaign was preregistered to measure it directly (`docs/campaigns/iqm_dla_window_variability_prereg_2026-07-22.md`). One nominal window is a complete 36-circuit execution on the pinned layout [2, 7, 12, 13] at depths 4, 8, 10 and 12, submitted as one main job and one readout job, with 4,096 shots per state and depth; the window is the unit of custody and of the frozen analysis. Ten windows were executed between 2026-07-22 and 2026-09-04.

Frozen primary (depth 10, window level). The even-minus-odd leakage difference is heterogeneous across the ten windows beyond shot noise: Cochran $Q = 19.01$ on 9 degrees of freedom, $p = 0.025$, with a between-window standard deviation $\tau = 0.0115$ against a mean shot-noise standard error of 0.0109 and a weighted mean difference of +0.0179. The preregistered rule "drift exceeds shot noise" is met (Fig. 3).

Secondary, by depth (Holm-adjusted). Homogeneity is rejected at depth 8 (adjusted $p = 4.4 \times 10^{-4}$) and not at depth 4 (0.053) or depth 12 (0.40). The between-window τ is 0.0107, 0.0176, 0.0115 and 0.0022 at depths 4, 8, 10 and 12. The depth-4 difference is positive in nine of ten windows (Clopper–Pearson [9] 95 % interval 0.55 to 1.00).

Calibration epochs (amendment, sensitivity analysis). While the campaign ran we found that several nominal windows carried the same provider calibration set: windows 2–3 and windows 7–10. Windows under one calibration set are technical replicates, not independent observations. An amendment (`docs/campaigns/iqm_dla_window_variability_epoch_amendment_2026-09-04.md`), committed before windows 8 to 10 were counted for inference, adds an epoch-pooled analysis and requires both results to be reported together. Pooled to six calibration epochs, the depth-10 heterogeneity is stronger, not weaker: Q = 15.55 on 5 degrees of freedom, $p = 0.0083$, $\tau = 0.0129$

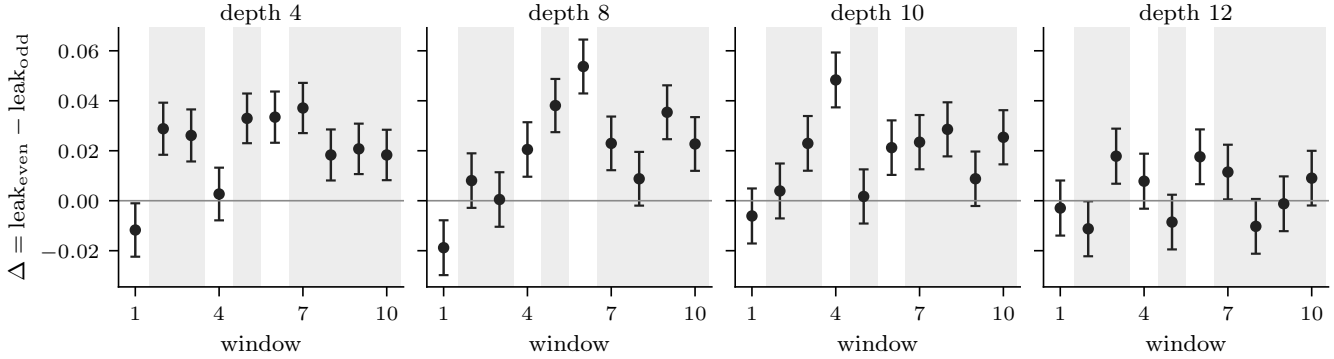


FIG. 3. Even-minus-odd leakage difference on Garnet in the ten execution windows of the window-variability campaign, by depth. Error bars are binomial shot-noise standard errors (4,096 shots per arm). Shading alternates with the provider calibration set: windows inside one band were taken under the same calibration set and are technical replicates.

against a shot-noise standard error of 0.0085. Homogeneity is rejected at depth 4 (adjusted $p = 0.012$) and depth 8 (5.6×10^{-5}), not at depth 12 (0.64; τ estimated at zero).

What this establishes. On this device and layout the parity-sector asymmetry at depths 4, 8 and 10 varies between calibration epochs by an amount comparable to or larger than the shot noise of a 4,096-shot arm. Adding shots inside one calibration window therefore narrows the wrong interval: the reported uncertainty of a single-window asymmetry understates how far the next window will be. This is why the depth-ordering question of Sec. V (one-sided $p = 0.064$ inside one calibration set) could not be settled by more shots, and it fixes the design of the confirmatory campaign that follows: the calibration epoch is the independent unit.

Boundary. A changed calibration-set identifier marks an operational epoch; it does not prove physical independence, and slow drift can cross calibration boundaries. No cause of the variability is claimed; the calibration covariates were preregistered as descriptive only.

VII. DEPTH-ORDERING CONFIRMATION ACROSS CALIBRATION EPOCHS

Design. Frozen before any of its data existed (`docs/campaigns/iqm_dla_depth_profile_powered_epoch_prereg_2026-09-04.md`). The six calibration epochs of Sec. VI gave the depth contrast $\mathrm{C} = (\text{leak}_{\text{even},8} - \text{leak}_{\text{odd},8}) - (\text{leak}_{\text{even},12} - \text{leak}_{\text{odd},12})$ a mean of 0.0164 and a between-epoch τ of 0.0150. Those values were used to choose the sample size and for nothing else: their counts and their six calibration sets are excluded from the confirmatory endpoint. The campaign consists of twelve distinct, previously unused calibration epochs. Each epoch runs the complete matrix in one pass: twelve repetitions of each of the four state-by-depth cells at 1,024 shots (12,288 shots per cell), plus four readout states at 2,048 shots — 48 main circuits in one provider job and 4 readout circuits in a second, 57,344 shots per epoch and 688,128 in total. An epoch is admitted only if its calibration set differs from all earlier ones and from the six design epochs, every earlier epoch has a complete paired calibration and count record, and the three edges of the layout exist in the snapshot. No circuit, layout, depth, repetition or shot count may change in response to a result; there is no optional stopping and no replacement of an admitted epoch.

Analysis (frozen). A DerSimonian–Laird [10] random-effects model of the twelve epoch contrasts with binomial within-epoch variances; one-sided test of H0: mean(C) ≤ 0 with the Hartung–Knapp–Sidik–Jonkman statistic [11, 12] on 11 degrees of freedom, its scale floored at one so that a small observed residual cannot narrow the interval [13, 14]. The null is rejected only if the estimated mean is positive and $p < 0.05$. Projected power at the design mean is 0.92. Raw parity leakage is primary; no readout correction is applied to the endpoint.

Execution. Twelve epochs between 2026-09-23 and 2026-10-06 on IQM Garnet, twelve distinct calibration sets, 24 provider jobs, 624 circuits. Every epoch passed the admission gate on its own day; the provider's payload digests equal the digests written to the submission journal before each call; the frozen validator accepted the counts of every epoch. The endpoint analysis was run once, after the twelfth epoch was validated, with the analysis script unchanged since its freeze on 2026-09-04 (`data/iqm_paper_replication/iqm_dla_depth_profile_powered_analysis_2026-10-11.json`).

Result. The null is rejected. The mean depth contrast is +0.0173 (standard error 0.0080; $t = 2.176$ on 11 degrees of freedom, one-sided $p = 0.026$), with between-epoch $\tau = 0.0261$ (Cochran $Q = 108.0$ on 11 degrees of freedom). The one-sided 95 % lower bound of the mean, from the same standard error and t distribution, is +0.0030. The contrast is positive in nine of twelve epochs and negative in three (epochs 1, 4 and 9). On this device and layout the even-odd leakage asymmetry is, on average over calibration epochs, larger at depth 8 than at depth 12: the ordering that Sec. V could not establish inside one calibration set is established across twelve (Fig. 4,

Table V).

The between-epoch τ is larger than the 0.0150 assumed when the sample size was chosen, so the campaign had less power than the planned 0.92; the rejection stands on the preregistered test as run, and the lower bound is close to zero. The result is a statement about the mean over epochs: single epochs differ by far more than their shot noise (per-epoch shot-noise standard error of the contrast about 0.0088), and in three of twelve the sign is reversed.

Secondary (preregistered). Random-effects summaries of the asymmetry at each depth separately: +0.0225 at depth 8 (standard error 0.0046, one-sided p = 0.00024, $\tau = 0.0146$) and +0.0051 at depth 12 (standard error 0.0056, $p = 0.19$, $\tau = 0.0183$). The six design epochs, which are not part of the endpoint, had contrasts −0.0159, +0.0010, +0.0127, +0.0466, +0.0361 and +0.0202. The four readout states of every epoch are committed with the counts and are not used in the endpoint.

Boundary. A statement about device noise on IQM Garnet, this layout, these two depths, this circuit family and this access period. Noiseless parity leakage of these circuits is exactly zero, so nothing here is a statement about coherent dynamics. No causal mechanism, no transfer to another backend, no advantage claim.

VIII. OUTLOOK: OTHER DEVICES AND PROCESSOR CLASSES

Everything reported here and in our earlier work was measured on two families of superconducting processors: heavy-hex IBM Heron r2 and square-lattice IQM Garnet. The pooled even-over-odd sign of the parity-sector leakage asymmetry is common to both; its depth dependence is not, and on Garnet it moves between calibration epochs by more than the shot noise. Two devices cannot tell whether these are properties of superconducting hardware in general, of particular two-qubit-gate implementations, or of these two machines.

The measurement is small enough to be repeated on any current gate-model processor: four qubits, number-conserving XY-Trotter circuits of depth 4 to 12, two initial states, and a few thousand shots per arm. Because the noiseless parity leakage of these circuits is exactly zero, the observable is a noise diagnostic that needs no classical reference beyond the counts themselves. Processor classes whose dominant error channels, connectivity and recalibration cadence differ from those of transmon devices — trapped ions, neutral atoms, spin qubits, and superconducting architectures with other couplers or lattices — are the informative comparison. We make no prediction for them: an asymmetry of the same sign, of the opposite sign, or a bounded null on another class would each constrain the explanation.

The protocol is published so that it can be run without us: the preregistration texts, circuit builders, admission checks, submission journals and frozen analysis scripts are in the public repository (Sec. IX). The one design lesson we would carry over is the unit of analysis: on Garnet the calibration epoch, not the shot, is the independent observation, and the equivalent recalibration event of another platform should be identified before the sample size is fixed. The confirmatory design used here costs twelve epochs of 57,344 shots. We welcome replications by device operators and independent groups, and would run the same preregistered protocol on other hardware where access is available.

IX. REPRODUCIBILITY

Everything needed to reproduce all five campaigns is committed to the public repository (github.com/anulum/scpn-quantum-control):

- preregistrations with frozen decision rules (and the amendment history);
- live calibration snapshot with calibration-set identifier;
- exact circuit payloads (QPY) and circuit-label manifests;
- submission records with raw IQM job identifiers (publishable; the access token is never committed);
- raw measurement counts for every reported hardware circuit;
- for the twelve-epoch campaign, the per-epoch submission journals with the digest of every circuit payload written before the provider call, and the admission check each epoch passed before it was submitted;
- frozen analysis scripts (fixed bootstrap seeds) whose committed artefacts contain every number quoted here;
- the harness modules with 100 % test coverage, hermetic against synthetic lattices.

X. CLAIM BOUNDARIES

No quantum-advantage claim (all circuits are classically simulable at these sizes — that is what makes the exact references possible). No coherence-protection claim. No coherent-dynamics claim for the parity asymmetry (noiseless leakage is exactly zero). No layout-transfer claim (the frozen rule failed). No extrapolation beyond the sampled device, calibration windows, chain lengths, depths, and schedules. The core results are statements about IQM Garnet on 2026-07-21 under calibration set `246a4930-54e3-4cd9-a2d1-fcc0919675f5`; the per-size study is separately bounded to its 2026-07-26 window under calibration set `c2097be4-1e23-49bc-adaa-8e8c01df6223`. The window-variability and depth-ordering results are statements about the pinned layout [2, 7, 12, 13] over the windows and calibration epochs listed in Secs. VI and VII; a distinct calibration set marks an operational epoch and does not prove physical independence, and the depth ordering is established for the mean over twelve epochs, not for an individual epoch.

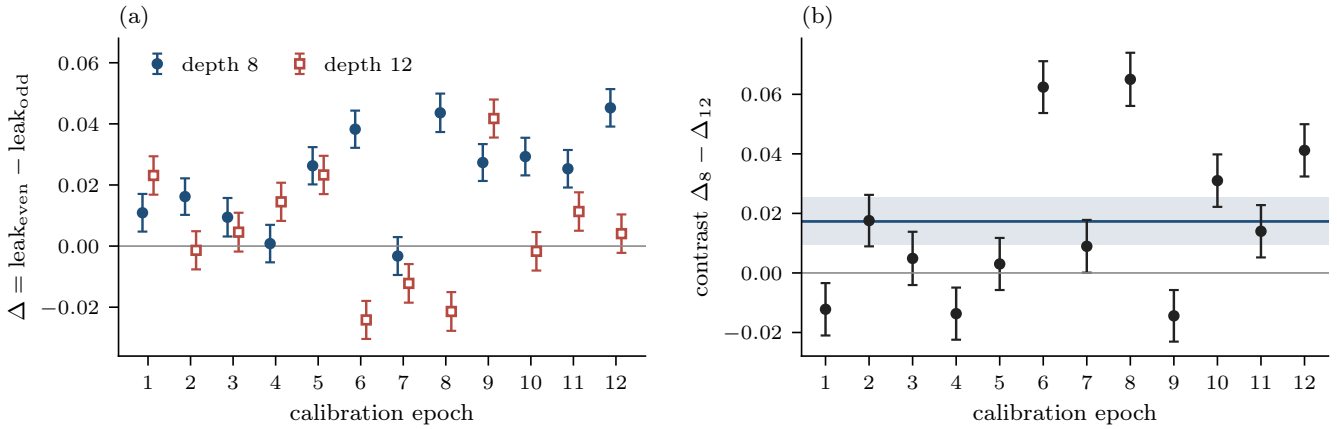


FIG. 4. Twelve-epoch depth-ordering confirmation on Garnet. (a) Even-minus-odd leakage difference at depths 8 and 12 in each calibration epoch (12,288 shots per arm). (b) The preregistered contrast $\Delta_8 - \Delta_{12}$ per epoch; the line and band are the random-effects mean and its standard error. Error bars are shot-noise standard errors and do not include the between-epoch spread.

epoch	date	calibration set	jobs (main, readout)	Δ_8	Δ_{12}	contrast
1	2026-09-23	`ad7e2bf0...`	`01a0cfbc-c583...`, `01a0cfbc-c711...`	+0.0109	+0.0231	−0.0122
2	2026-09-24	`7c39177a...`	`01a0d203-4e04...`, `01a0d203-4fd7...`	+0.0162	−0.0014	+0.0176
3	2026-09-25	`af33e1ea...`	`01a0d916-2e1a...`, `01a0d916-2f53...`	+0.0094	+0.0046	+0.0049
4	2026-09-26	`d10715c4...`	`01a0dec6-b2b3...`, `01a0dec6-b3f0...`	+0.0008	+0.0145	−0.0137
5	2026-09-27	`e1084b22...`	`01a0e410-5e40...`, `01a0e410-5f79...`	+0.0263	+0.0233	+0.0030
6	2026-09-29	`a1a74f13...`	`01a0ef34-e362...`, `01a0ef34-e611...`	+0.0382	−0.0242	+0.0624
7	2026-10-01	`36cf327c...`	`01a0f933-9f69...`, `01a0f933-a1de...`	−0.0033	−0.0122	+0.0090
8	2026-10-02	`a4b99350...`	`01a0fbf5-baa8...`, `01a0fbf5-bc08...`	+0.0436	−0.0214	+0.0650
9	2026-10-03	`0660d260...`	`01a101dc-eada...`, `01a101dc-ec11...`	+0.0273	+0.0417	−0.0144
10	2026-10-04	`e431ada3...`	`01a10729-ba87...`, `01a10729-bd8d...`	+0.0293	−0.0017	+0.0310
11	2026-10-05	`ff6836ae...`	`01a10da8-ca46...`, `01a10da8-cbf2...`	+0.0253	+0.0113	+0.0140
12	2026-10-06	`6dd7fae6...`	`01a11091-86f0...`, `01a11091-890d...`	+0.0452	+0.0041	+0.0412

TABLE V. Twelve-epoch depth-ordering confirmation. Δ is the even-minus-odd leakage difference at the given depth, pooled over the twelve repetitions of the epoch; the contrast is $\Delta_8 - \Delta_{12}$. Calibration-set and provider job identifiers are abbreviated; the full identifiers are in the committed analysis artefact and submission journals.

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